a

ENS3553/ENS6160 Signals and Systems

Laboratory Experiment 2

Frequency Response of Systems – Lab Results

Student Name:_____ Student No:_____

Post-lab

A. Frequency response of an RC circuit

a) Determining the frequency response,

b) Calculating | |

f	(rad/s)	| |	| | in dB
1 kHz	$= 2\pi(1000)$ $= 6283.19$ rad/s	0.9956	-0.0384
10 kHz	62831.85	0.7277	-2.76
40 kHz	251327.41	0.2564	-73.71

c) Plot of amplitude and phase response

nd Systems

Laboratory Experiment 2

Frequency Response of Systems – Lab Results

Student Name:_____ Student No:_____

Post-lab

A. Frequency response of an RC circuit

a) Determining the frequency response,

f	ω(rad/s)	$\lvert H(j\omega)\rvert$	$\lvert H(j\omega)\rvert$ *in dB*
1kHz	6283.19	0.9956	-0.0384
10kHz	62831.85	0.7277	-2.76
40kHz	251327.41	0.2564	-11.82

b) Calculating |

c) Plot of amplitude and phase response

d) Questions

a) What kind of filtering is performed by this RC circuit?

b) What does -3db equate to in terms of voltage amplitude ratio?

c) What is the cut-off frequency (-3dB point on the plot)?

d) How can we determine this value from the circuit parameters?

e) Describe the behaviour of this system in the frequency domain and in the time domain.

B. Response of the RC circuit to an AC voltage using the frequency response

a) Theoretical output voltage

f	(rad/s)	| |	∠	| |	∠
1 kHz	6283.19	0.9956	-5.38	7.467	-5.38
10 kHz	62831.85	0.7277	-43.30	5.458	-43.30
40 kHz	251327.41	0.2564	-75.14	1.923	-75.14

f	ω(rad/s)	$\lvert H(j\omega)\rvert$	$\angle H(j\omega)$	$\lvert Vo\rvert$	$\angle Vo$
1kHz	6283.19	0.9956	-5.39°	14.93	-5.39°
10kHz	62831.85	0.7277	-43.30°	10.92	-43.30°
40kHz	251327.41	0.2568	-75.14°	3.85	-75.14°

b) Magnitude and phase spectra of output voltage

C. Response of the RC circuit to square wave using the frequency response

a) Fourier Series coefficients

$=$

$=$

$$f_0 = 8000 \text{ Hz}$$

Hz

Fourier Coefficient(b_n)	Value	Frequency
b1	9.55 V	8kHz
b3	3.18 V	24kHz
b5	1.91 V	40kHz

(b) and (c) Magnitude of frequency response and Magnitude of harmonics

f (Hz)		| |	Input magnitude (			
			V	dBV	V	dBV
Fundamental =						
3rd harmonic =						
5th harmonic =						

***f*(Hz)**	***ω*(rad/s)**	***|H(jω)|***	Input magnitude(b_n)		Output magnitude	
			V	dBV	V	dBV
Fundamental=8000Hz	50265. 48	0.7985	9.55	19.60	7.63	17.65
3rd harmonic=2000Hz	150796. 45	0.4041	3.18	10.06	1.29	2.19
5th harmonic=4000Hz	251327. 41	0.2564	1.91	5.62	0.49	-6.19

d) Description of output waveform

Results Sheet

A. Verify frequency response in time domain

1. Table A: Theoretical frequency response vs measured frequency response

Frequency	**Magnitude**		**Phase**	
	Theoretical Value	**Measured Value**	**Theoretical Value**	**Measured Value**
1 kHz				
10 kHz				
40 kHz				

Frequency	Magnitude		Phase	
	Theoretical Value	Measured Value	Theoretical Value	Measured Value
1kHz	14.93	15.20	-5.39°	-6.65°
10kHz	10.92	11.50	-43.30°	-42.30°
40kHz	3.85	4.22	-75.14°	-73.71°

2. Sketch of input-output pairs.

3. Filtering type and comparison of cut-off frequency (-3dB point) between the measured and theoretical values.

B. Verify frequency response in frequency domain

1. Table B: Frequency response (magnitude) for sinusoids

Frequency	Output Magnitude	Input Magnitude	Magnitude Gain	Theoretical Response
1 kHz	dBV	dBV	dB	dB
10 kHz	dBV	dBV	dB	dB
40 kHz	dBV	dBV	dB	dB

Frequency	Output Magnitude	Input Magnitude	Magnitude Gain	Theoretical Response
1kHz	14.61	14.83dBV	-0.23dB	-0.0384dB
10kHz	12.18dBV	14.99dBV	-2.81dB	-2.7606dB
40kHz	3.47dBV	14.83dBV	-11.35dB	-11.821dB

2. Comparison of measured results with theoretical (post-lab) values

3. Sketches of output spectra and description of comparison

C. Response to square wave input

1. Table C: Square wave theoretical vs measured harmonic response

Spectrum characteristic	**Output Magnitude**	**Input Magnitude**	**Magnitude Response**	**Theoretical Response**
Fundamental	dBV	dBV	dB	dB
3rd harmonic	dBV	dBV	dB	dB
5th harmonic	dBV	dBV	dB	dB

Spectrum characteristic	Output Magnitude	Input Magnitude	Magnitude Response	Theoretical Response
Fundamental	20.19dBV	22.14dBV	-1.95dB	-1.96dB
3rd harmonic	4.72dBV	12.60dBV	-7.87dB	-7.87dB
5th harmonic	-3.65 dBV	8.17dBV	-11.82dB	-11.82dB

2. Sketch of output spectra

3. Sketch of input and output time domain waveforms and explanation

Discussion of experimental results

~ End of Lab Report ~